SIMSHIELD

Network-native identity verification & continuous KYC fraud prevention

GSMA Open Gateway APIs · Nokia NaC Platform · Google Gemini AI

Open Gateway Hackathon 2026 · Talent Arena

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& continuous KYC fraud prevention

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Nokia Network as Code · GSMA Open Gateway

"Every fraud attack starts with a SIM card nobody checked."

SIMShield makes the telecom network
the identity layer it was always meant to be.

SIM Swap	**KYC Match**	**Device Swap**	**Device Location**

4 LIVE APIs · Nokia NaC · Movistar · Orange · Vodafone

< 300ms	**DORA Ready**	**Zero-Data**	**3 Operators**
avg latency	EU 2025	GDPR-native	~95% Spain

Presented by: Michella (Product Owner) · Victor (AI/Tech) · Armando (PoC/Dev)

THE PROBLEM — IDENTITY FRAUD IS A SYSTEMIC RISK. CURRENT DEFENCES WERE NOT BUILT FOR THIS.

Business Relevance: Real persona · Concrete consequences · Why acting now protects operators and clients

THE SCENARIO

María is shopping in Madrid. 450km away in Seville, a fraudster tries to use her cloned card at a POS terminal.

The bank fires a verification request to SIMShield. We check via Device Location API: SIM swapped 38 minutes ago. Device in Seville. Card registered in Madrid. Score: 9.

SIMShield's chatbot alerts María: "Are you in Seville?" She replies: "No."

Transaction denied. Fraud stopped.

£300M

M&S fraud loss

Outsourced helpdesk SIM swap
Q1 2025 — in this room's sector

1%

SIM swap surge

UK 2023→2025. Spain following.
Fraud ring industrial scale.

€2.1B

Card-present EU fraud

POS terminal attacks.
Phone 450km from card.

WHY CURRENT DEFENCES FAIL

SMS 2FA

Bypassed when swap completes.
OTP routes to attacker.
Bank has zero visibility.

Bank Fraud AI

Fires after transaction.
SIM swap happened minutes before — outside bank window.

Static KYC

Checked once at onboarding.
Never re-verified when SIM changes hands mid-relationship.

Carrier PINs

Outsourced helpdesk bypass under pressure. M&S exact vector. Not scalable.

FOR OPERATORS — PROACTIVE FRAMING: *Reducing fraud proactively lowers chargeback liability, reduces regulatory pressure, and protects operator brand — before AEPD enforcement. Fine exposure and regulatory detail are out of scope today but have been modelled — ask us in Q&A.*

PoC ARCHITECTURE — 4 LIVE APIs · NOKIA NaC · sim-shield.vercel.app · < 300ms

Technical Demonstration: Live on Vercel · Nokia NaC platform · 3 operators · GSMA CAMARA standard

SIMShield verifies the SIM, not just the password — before every transaction, in real time, on all 3 Spanish operators.

USER LAYER

Merchant POS · Bank App · Enterprise Helpdesk · ADP HR Platform

SIMSHIELD PRODUCT LAYER

FastAPI backend · Gemini AI Agent · MCP Protocol · Vercel deployment · React dashboard + Mapbox

NOKIA NaC PLATFORM

network-as-code.developer.nokia.com · GSMA CAMARA s

SIM Swap

Swapped last 24h?

or

KYC Match

Name/DOB vs live record

Device Swap

Hardware changed?

Device Location • LIVE

Phone at POS? Delta calc.

OPERATOR NETWORK LAYER

Movistar · Orange · Vodafone — 3 live 5G networks — ~95% Spanish mobile subscribers

TECH STACK

Backend: Python FastAPI **Deploy:** Vercel **APIs:** Nokia NaC SDK **AI:** Gemini + MCP **Frontend:** React + Mapbox **Pro**

• sim-shield.vercel.app

287ms

avg latency per check

AI INTEGRATION + SECURITY & COMPLIANCE — GEMINI · MCP · ZERO-DATA BROKER · DORA READY

Technical Innovation: Agentic AI for real decisions · Human-in-loop · GDPR-native · Financial risk reduction

MCP TOOLS

- `check_sim_swap()`
- `kyc_match()`
- `device_swap()`
- `check_location()`

DECISION CHAIN

TRIGGER
Payment · Login · Helpdesk

02 PARALLEL CALLS
4 MCP tools via Nokia NaC

03 AI REASONING
Gemini chain-of-thought

04 SCORE 0–100
SIM 30% · KYC 25% · Loc 20% · Dev 15%

05 DECIDE / A2A
<30 approve · 30-70 escalate · >70 block

GEMINI VERDICT — DEMO MOMENT

"SIM swapped 38 min ago.
Device 450km from POS.
KYC mismatch confirmed.
Confidence: 97%.

Recommendation: BLOCK."

BLOCKED — 97% CONFIDENCE

SECURITY & GDPR

- Zero raw data on SIMShield servers — hashed tokens only
- Gemini sees signals only — never MSISDN, name, coordinates
- TLS 1.3 · Cloud Secret Manager · GDPR Art. 6(1)(f)

COMPLIANCE & FINANCIAL RISK REDUCTION — WHY OPERATORS & MNCs ACT NOW

DORA (EU 2025)

Real-time identity controls mandated. SIMShield is compliant by design.

PSD3 / SCA

Stronger auth required. SIM verification is the missing layer.

AEPD Awareness

Operator fine risk noted. Out of scope today — ask us in Q&A.

MNC Risk (ADP)

Helpdesk impersonation = payroll fraud vector. M&S blueprint.

Operator ROI

Lower chargebacks, lower regulatory risk, 15% rev-share.

COMMERCIAL VIABILITY — PHASED FINANCIAL MODEL · CLIENT TIERS · ROI

Commercial Viability: Business model · Target market · Value proposition · Phase 1 MVP → Phase 2 MVT → Phase 3 Scale

TARGET CLIENT SEGMENTS

BEACHHEAD — MVP

Spanish Neobanks & Fintechs

Rebellion Pay · Bnext · Revolut ES · N26 ES

€0.004/call

SCALE — PROCESSORS

Payment Processors

Redsys — 1 contract = national coverage

Rev share

ENTERPRISE MNC — MVT

Large Corporates + HR Platforms

ADP · Workday · SAP SuccessFactors · BBVA · Santander

€8/seat/mo

SME — HOLDED/QUIPU

Spanish SMEs via Plugin

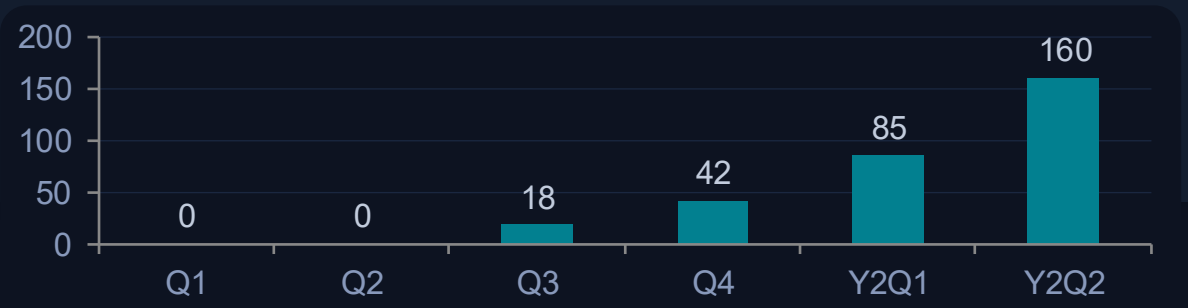
200K addressable · Holded + Quipu ecosystem

€29/mo

TAM	SAM	SOM
$47B	$8.4B	$220M

PHASED FINANCIAL MODEL

	PHASE 1 MVP Now → Q2 2026	PHASE 2 MVT Launch Q3 2026→Q1 2027	PHASE 3 Scale Q2→Q4 2027
Scope	PoC · demo day		20 fintechs + 2 MNC pilots
Monthly checks	Live demo		~2M / month
ARR	€0		~€85K
Cost / check	€0.036 avg		€0.036 avg
Break-even	—		Month 8–10
MNO rev share	—		15% to operators
Spain API $$	Not monetised		Launching 2026
Key milestone	Hackathon win		Redsys integration



NOKIA + TELEFÓNICA TESTBED vs SIMSHIELD — 4 SPECIFIC GAPS

Dimension	Nokia + Telefónica (Lab)	SIMShield (Product)
APIs used	SIM Swap + Device Swap (2 APIs)	SIM Swap + KYC Match + Device Swap + Device Location (4 APIs)
Operators	Telefónica only — single MNO	Movistar + Orange + Vodafone — ~95% Spain
KYC signals	None	KYC Match: live name/DOB vs operator record
Geolocation	None — digital fraud only	Device Location vs POS — card-present fraud stopped
AI layer	A2A + MCP protocol testing	Gemini making real decisions · scored output · human-in-loop
Commercial	Lab PoC, no pricing	€0.004/call · €29/mo SME · €8/seat MNC · deployable today
Compliance	Not addressed	DORA-ready · zero-data broker · GDPR Art. 6(1)(f)

SCALABILITY

- GSMA CAMARA → any operator globally
- Nokia NaC SDK → single integration, multi-MNO
- Cloud-native stateless API → 50M checks/mo
- ADP/Workday → 1 HR integration = all employees

ROI SUMMARY

1 incident prevented pays for:	> 116,000 checks
Cost per check:	~€0.036 incl. SMS
ROI per prevention:	> 2,000×

NOKIA BUILT THE PIPES. SIMSHIELD BUILT THE PRODUCT. THE APIS ARE LIVE. THE OPERATORS ARE IN THIS ROOM.

That product is SIMShield.

SIMSHIELD

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SIMShield makes the telecom network the identity layer it was always meant to be. Every day, banks approve transactions based on a PIN and a password. What they don't check is whether the phone number on that account still belongs to the same person. We built SIMShield to close that gap.

SIMShield makes the telecom network the identity layer it was always meant to be.

Third-party system fraud protection

SIMSHIELD

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Our API conectors

4 APIs · 3 operators · <300ms fraud verdict

- SIM Swap
- KYC Match
- Device Location
- Device Swap

SIMSHIELD

Differentiator 1
KYC signal stack Nokia/Telefónica testbed uses SIM Swap and Device Swap. You use 8 APIs, including the full KYC family: Match, Tenure, Age, and Fill-in. KYC Tenure specifically — how long has the SIM been active — is not in their testbed and is one of the strongest fraud signals available. A 2-day-old SIM on a 10-year-old bank account is a fraud ring signature. Their system cannot see that. Yours can.

Differentiator 2
Multi-operator aggregation Their testbed is Telefónica only. SIMShield runs across Movistar, Orange and Vodafone simultaneously. In Spain, that means ~95% of mobile subscribers are covered by one API call. A fraud check that works only with one operator is not commercially viable. Nokia NaC is the aggregation layer that makes this possible

Differentiator 3
Physical payment geolocation Their use case is purely digital fraud — SIM swap for account takeover. SIMShield adds the physical payment layer: Location Retrieval API compared against POS terminal coordinates. This is a completely new use case that their testbed does not cover. Card-present fraud at physical terminals is a €2.1B annual problem in Europe that no network-layer product addresses today.

Differentiator 4
Commercial product with SME pricing , Nokia and Telefónica are building infrastructure and protocols. They are not building a €29/month Holded plugin for Spanish SMEs. They are not building a DORA compliance dashboard for bank KYC officers. They are not building an enterprise helpdesk verification layer. The commercial product and go-to-market is entirely yours. This is the feasibility argument — you have a path to revenue that their initiative explicitly does not pursue.